

- 6 The curve C has polar equation $r = a \tan\left(\frac{1}{8}\theta\right)$, where a is a positive constant and $0 \leq \theta \leq 2\pi$.
- (a) Sketch C and state, in terms of a , the greatest distance of a point on C from the pole. [3]

- (b) Find, in terms of a , the area of the region bounded by C and the initial line. [4]


$$4 \sin\left(\frac{1}{4}\theta\right) \cos \theta + \sin \theta = 0$$

[6]

[illegible]